QP CODE:25900118

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCIAM101
AUTOMOTIVE SYSTEMS FOR E - VEHICLES
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours. **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions

Answer all questions

Each question carries 1 marks

1	**A Unity Gain Amplifier using an op-amp is also known as** Ans. Voltage follower	[K] / [CO1]
2	**What is commonly added to the output of a bridge rectifier to smooth the DC signal?** Ans. capacitor	[K] / [CO2]
3	**What does BLDC stand for?** Ans. Brushless DC	[U] / [CO3]
4	**In a photodiode amplifier circuit, the operational amplifier configuration is** Ans. Transimpedance amplifier	[U] / [CO1]
5	**Which sensor is commonly used in ESCs to monitor the motor's current and voltage in real time?** Ans. Current and voltage sensor	[U] / [CO3]
6	**In an inverting amplifier, which of the following resistors determines the gain of the amplifier?** Ans. Both Rin and Rf	[K] / [CO1]
7	**In a comparator circuit using the 741 op-amp, if the reference voltage (inverting input) is set to 0V, what would be the output state when the non-inverting input exceeds 0V?** Ans. Positive saturation	[K] / [CO2]
8	**What is the primary function of an Electronic Speed Controller (ESC) in an Electric Vehicle (EV)?** Ans. To regulate the speed of the motor	[U] / [CO3]
9	**What is the primary function of a 78XX voltage regulator IC?** Ans. To provide a constant output voltage	[U] / [CO2]

10 **The rotor current in an induction motor is induced by:** [U] / [CO3]

Ans. The magnetic field produced by the stator

[1x10 = 10]

Part B

Short Answer Questions

Answer any 4 questions

Each question carries 5 marks

11 **Compare single, dual and quad packages.** [U] / [CO2]

For 5 comparisons – 5 Marks

12 **Briefly explain the working of Schmitt Trigger.**

Diagram – 3 Marks [U] / [CO2]

Explanation – 2 Marks

13 **Explain the basic structure of a lithium-ion battery**.

Diagram – 3 Marks [K] / [CO2]

Explanation – 2 Marks

14 **Describe the symbol of an operational amplifier, its components, and explain the significance of each part in the circuit.**

Symbol – 2 Marks [U] / [CO1]

Components – 1 Mark

Significance – 2 Marks

15 **What role does the switching transistor play in an SMPS?** [U] / [CO2]

Explanation - 5 Marks

16 **What is DC fast charging, and why is it important?**

DC Fast Charging - 3 Marks [U] / [CO3]

Importance - 2 Marks

[5x4 = 20]

Part C

Essay Questions

Answer any 2 questions

Each question carries 10 marks

17 **Explain open loop op amp configurations.**

Explanation - 6 Marks [U] / [CO1]

Diagram – 4 Marks

18 **Define a Battery Management Unit (BMU). Describe its primary functions, components, and their roles in ensuring safe and efficient battery operation.**

Definition - 2 Marks [U] / [CO3]

Primary Function - 3 Marks

Components - 3 Marks

Role – 2 Marks

19 **Discuss the effect of a capacitor filter in a bridge rectifier circuit.**

Diagram – 4 Marks

Explanation of Capacitor Filter – 2 Marks

Explanation for its effect in bridge rectifier – 4 Marks

[U] / [CO2]

20 **Describe the various types of EV charging connectors used globally. Discuss their specifications and compatibility with EVs.**

Description of various types -5 Marks

Discussion of their specifications and compatibility with EVs. – 5 Marks

[K] / [CO3]

[10x2 = 20]